



WESTERN SYDNEY
UNIVERSITY

Assessment 2 (30%)

Applications of Big Data (Comp3002)

Scalable and Explainable Sentiment Analysis System

Due Date: 07-06-2026

Overview

In this assignment, you will design and implement a sentiment analysis system that works on large-scale movie review data.

Your system should not only predict whether a review is positive or negative, but also:

- scale to large datasets using distributed computing
- explain its predictions
- handle unclear or mixed (ambiguous) reviews

This assignment focuses on solving a complex computing problem where there is no single correct solution.

Learning Objectives

- Apply distributed computing (MapReduce) for large-scale data processing
- Build and evaluate machine learning models
- Explain how your model makes predictions
- Handle ambiguous and real-world data challenges
- Justify technical decisions in system design

Dataset

You will use the IMDB movie review dataset provided on vUWS.

- 25,000 training reviews
- 25,000 testing reviews
- /train/pos – positive training reviews
- /train/neg – negative training reviews
- /test/pos – positive test reviews
- /test/neg – negative test reviews

Task 1: Distributed Feature Engineering (10 marks)

- Use Hadoop MapReduce to compute TF-IDF features
- Convert reviews into numerical vectors
- Implement one alternative feature method
- Compare performance and scalability

Task 2: Classification Model (10 marks)

- Build a sentiment classification model
- Use at least two classifiers
- Perform parameter tuning

- Compare performance and justify your choice

Task 3: Explainability (5 marks)

- Identify important features influencing predictions
- Provide explanations for predictions

Task 5: Handling Ambiguous Reviews (5 marks)

- Identify ambiguous reviews
- Propose and explain a handling method

Implementation Requirements

- Use Python
- Use Hadoop Streaming
- Use scikit-learn
- Ensure clean, well-documented code

Assessment Submission

Your submission should include:

1. Source code and Hadoop MapReduce implementation
2. Any scripts/notebooks used for preprocessing, feature engineering, training, and evaluation [**Note: Both 1 and 2 could be in same Jupyter notebook file**]
3. A short report (approximately 1-2 pages) discussing:
 - What sentiment analysis is? Why large-scale text processing is important?
 - the overall approach and technical decisions
 - feature engineering and classifier selection
 - model performance and comparison of results
 - explainability of predictions
 - handling of ambiguous or mixed-sentiment reviews
 - challenges, limitations, and possible improvements

Students are expected to justify their design choices and demonstrate understanding of distributed computing, machine learning, and real-world sentiment analysis challenges.

Overall Marking Criteria

Your submission will be assessed against both the quality of the written/report component and the technical implementation. The marking criteria are designed to evaluate how well your work addresses the task requirements, explains the adopted approach, and demonstrates correct, usable, and maintainable implementation.

The assessment will consider the following areas:

1. **Purpose, Structure and Coverage:** clarity of purpose, completeness of required sections, and logical organisation.
2. **Approach / Method Explanation:** quality of explanation, justification of decisions, and demonstrated understanding.
3. **Referencing and Formatting:** accuracy of referencing, consistency of style, and professional presentation.
4. **Correctness and Completeness:** extent to which the solution meets the task requirements and produces correct outputs.
5. **Code Structure and Maintainability:** readability, modularity, organisation, naming conventions, and avoidance of unnecessary duplication.
6. **Testing / Validation and Usability:** evidence of testing, reliability, clear instructions, input validation, and ease of use.